# Chamudi De Silva

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## RESEARCH INTERESTS

Computational Cognitive Science, Computational Linguistics, LLM Evaluation, Pragmatics, and Human-Centered AI

## EDUCATION

**University of Peradeniya, Peradeniya, Sri Lanka** 2021 – 2025
*BSc. (Hons) in Computer Science, minor in Statistics*
CGPA: 3.55/4.00 — Second Class Honours (Upper Division)
GPA for most recent 60 graded credits: 3.73/4.00

**Selected coursework**

- *Computer Science*: Artificial Intelligence, Neural Networks and Deep Learning, Machine Learning, Design and Analysis of Algorithms, Distributed Systems
- *Statistics*: Statistical Data Mining, Sampling Techniques, Probability Theory, Non-Parametric and Categorical Data Analysis

## RESEARCH EXPERIENCE

**Ongoing Projects** Ongoing

- Physics-Informed Machine Learning for History-Dependent Hemodynamic Responses in fMRI
  *Supervisors: Dr. Ruwan Ranaweera, Prof. Amalka Pinidiyaarachchi*
- EEG Correlations of Creativity, Personality, and Vocational Interest
  *Supervisors: Prof. Roshan Ragel, Dr. Ruwan Ranaweera, Dr. Erunika Dayaratna*
- Toward Ethnically Equitable Emotional Intelligence in LLM-Based Agents
  (Extending the Shades of Sentiment study (accepted, ISBM 2026))
  *Supervisor: Dr. Erunika Dayaratna*

**A Multi-Dimensional Study of Combined XAI Techniques for Regression** Jun – Sep 2026
*Supervisor: Dr. Erunika Dayaratna*

- Evaluated single and combined XAI methods (SHAP, DiCE, Anchors, ProtoDash) on faithfulness, actionability, stability, and consistency.
- Found that well-chosen combinations improve several criteria, depending on model and aggregation strategy.
- Built a prototype that turns combined explanations into audience-tailored natural-language guidance.

**Evaluating Emotional Fairness Across Ethnic Identities in LLMs** Feb – Jun 2026
*Supervisor: Dr. Erunika Dayaratna*

- Audited seven LLMs on emotion-conditioned character profiles, using fairness metrics such as Demographic Parity Difference.
- Found skew toward Male and Global North personas and emotional and occupational stereotyping in every model.
- Showed that repeated prompting amplified bias by driving models toward near-deterministic outputs.

**Integrating CL and RL for Personalised Response Generation in Companion Agents** Mar – Dec 2025
*Supervisor: Dr. Ruwan Nawarathna*

- Proposed a hybrid continual and reinforcement learning framework combining user persona modelling, memory adaptation, and RL-based response selection.
- Built a companion agent as an evaluation platform to test personalised, emotion-aware responses.
- Showed improved user-specific adaptation and more emotionally aligned responses.

## MANUSCRIPTS

**Shades of Sentiment: Evaluating Emotional Fairness Across Ethnic Identities in LLMs (Presented)** 2026
W. Kekulandara, **C. De Silva**, E. Dayaratna.
*5th World Conference on Information Systems for Business Management – ISBM*

**Is More Explanation Better? A Multi-Dimensional Study of Combined XAI Techniques for Regression (Accepted)** 2026
**C. De Silva**, W. Kekulandara, E. Dayaratna.
*8th IEEE International Conference on Advancements in Computing – ICAC*

**Hybrid Continual and Reinforcement Learning Framework for Adaptive Intelligent Companionship (In Prep)** 2026
**C. De Silva**, R. Nawarathna

## TEACHING EXPERIENCE

**Teaching Assistant** Jan 2026 – Present
*Dept. of Statistics and Computer Science, Faculty of Science, University of Peradeniya*

- Conduct weekly programming laboratories for 200+ undergraduate students.
- Teach introductory C and Python.
- Deliver substitute lectures in Operating Systems Concepts.
- Develop/evaluate instructional materials and assessments.

## TECHNICAL SKILLS

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| **Programming Languages** | Python, R, SQL |
| **ML & Data Analysis** | PyTorch, scikit-learn, NumPy, Pandas, SciPy, Matplotlib |
| **Databases & Tools** | MySQL, VS Code, Google Colab, LaTeX |

## LEADERSHIP & CO-CURRICULAR ENGAGEMENT

**Computer Society University of Peradeniya (CSUP)** 2024 – 2025
*Editor*
Editorial oversight of student submissions and event documentation.

**IEEE Computer Society Student Chapter, University of Peradeniya** 2024 – 2025
*Inaugural Assistant Secretary*
Supported establishment and coordination of the university chapter; oversight of editorial operations.

**Student Academic Advisory Committee (SAAC)** 2022 – 2026
*Organising Committee Member*
Co-organisation of the workshop series and multi-day official orientation programmes for incoming students of Faculty of Science.

**Leo Club of University of Peradeniya** 2023 – 2024
*Chief Editor*
Oversight of all editorial operations; management of a trilingual editorial team by supervising group members.

## LANGUAGES

**English** (fluent) · **Sinhala** (native) · **Tamil** (reading)

## PERSONAL INTERESTS

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| **Reading** | Contemplative poetry (Mary Oliver), fiction |
| **Languages** | Learning Spanish and Korean |
| **Other** | Travelling |